

Maṇḍala literalism

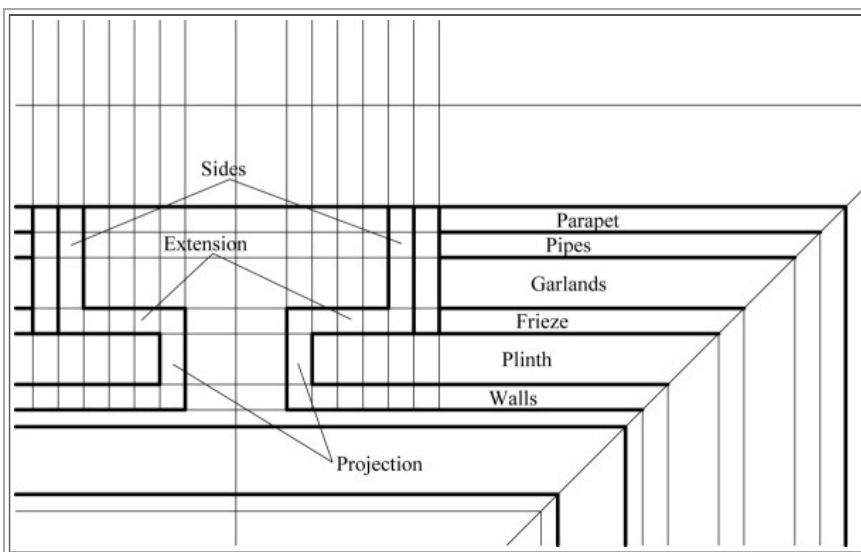
The information that comes to us today from the Tibetan traditions regarding the drawn 2D (*leb thig*) and constructed or imagined 3D (*blos bslangs*) Kālacakra maṇḍalas is very extensive, in both the Gelug and Jonang traditions; particularly the latter, with the pedantic attention to detail of Banda Gelek ('*ba' mda' dge legs*). With the 3D maṇḍala there is some room for artistic choice in many areas, but most of the architectural details are clearly described in terms of shape, size and colour. So, what is the source of this information?

From the original Indian traditions, there is a brief description of the 2D drawing in the Vimalaprabhā, the main Kālacakra Tantra commentary. In addition there are two notable texts with more extensive descriptions: the [Vajrāvalī](#) (*rdo rje'i 'phreng ba*) of Abhayākara Gupta and the maṇḍala ritual (*dpal dus kyi 'khor lo'i dkyil 'khor gyi cho ga*) by Sādhuputra. These are not as extensive as the later Tibetan texts, but are clearly the main textual sources that the Tibetan writers used. Of the 3D maṇḍala, there are scattered comments here and there, but no systematic point-by-point description as we find in the Tibetan traditions – and as there exists in the Indian texts for the 2D drawing. Clearly, the Tibetan understanding came from the oral traditions, and a certain amount from interpretation. This article discusses the interpretation of the Jonang writer [Chokle Namgyal](#) (*phyogs las nam rgyal*). He was a student of the great Dolpopa, and was also the first Kālacakra teacher of Tsongkhapa, the founder of the Gelug tradition.

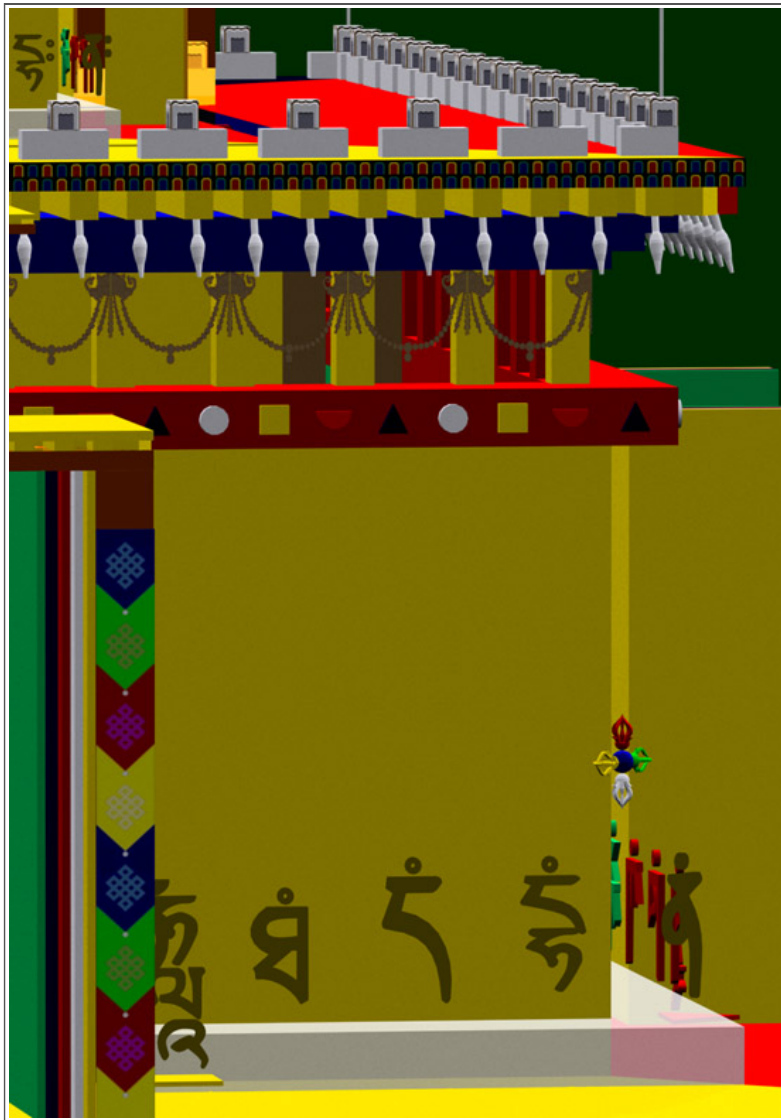
When interpreting the 3D structure of the Kālacakra maṇḍala palace, in this writer's opinion he fell victim to an unfortunate literalism that would almost certainly not have met with the approval of the early Indian Kālacakra masters, but in doing so he produced a very beautiful image and described a couple of interesting insights along the way. The following is based on his text: *dpal dus kyi 'khor lo'i bsgom bya'i dkyil 'khor phyi nang kun gsal* (pdf: [available here](#)).

A 2D maṇḍala, whether of Kālacakra or some other deity, is a two-dimensional representation of a three-dimensional structure. In the associated meditation practice, the practitioner imagines being the particular deity, inside the maṇḍala palace, together with a retinue, which in the case of Kālacakra numbers over 600 other deities. The 2D drawn maṇḍala cannot possibly include all the details of the 3D structure and necessarily involves considerable compromise in representing its features. It is essentially a floor plan of a multi-storied building.

Consider the drawing at the top of this page. This shows part of the body palace (lowest storey) in the Kālacakra maṇḍala. The narrow parallel coloured lines – green, black, red, white and yellow – are the walls of the palace. These are five narrow walls standing almost touching each other, and the lines drawn show us exactly where these walls stand, just as in a modern floorplan. The drawing can give us no information about the height of these walls, and it surely would be quite misleading to try and interpret their height from any feature that can be found in the drawing. The inside of the palace is below the horizontal lines of the walls as seen here, and above these, the walls run forwards to form a porch.



The first part of the porch is known as the porch-projection (*sgo khyud*, *niryāha*). This is a passage in the middle of which, in a 3D structure, would be placed the door itself – two door leaves (*sgo glegs*) enclosed within a door-frame (*sgo ru*). This component is not even hinted at in a 2D drawing. The distance between the parallel walls of the projection is taken as a basic unit of measurement, known as a door-unit (DU, *sgo tshad*). The internal length of the projection is one DU and the overall thickness of the walls is $\frac{1}{4}$ DU. The porch-extensions (*sgo 'gram*, *kapola*) then widen the structure to create the porch itself, running forwards in the form of the porch-sides (*sgo logs*, *pakṣaka*). The internal length of the walls of the projections, extensions and sides are all one DU. (The usage of these terms here is the normal one in a Kālacakra context, but sometimes the sides and projections are interchanged. That is partly why these particular English words have been used here as translations – interchanging them does not create any



particular problem.)

Immediately left and right of the porch-sides are represented the pillars that support the toran above the porch. These have a length of $1\frac{1}{4}$ DU. In the image above, these have coloured pennants down them as pillar decorations, in orange, blue, red and green. These pillars are not labelled in the second image; that image also shows some of the construction lines (*skam thig*) used in drawing the maṇḍala; these are more faint than the final lines (*rlon thig*). We will return to these pillars shortly, but before doing so it is worth describing some of the other features seen in these images.

The plinth (*stegs bu*, *vedika*, also '*dod snam*') is either a base for a wall, that protrudes beyond the bottom of the wall, or a course running around the bottom of a wall giving the impression of a base on which the wall sits. In a Kālacakra context it is usually considered to be the latter (and is white). Sitting on top of the wall is the red jewelled frieze (*rin chen snam bu/pha gu*, *ratnapaṭṭika*) running around the length of the wall and overhanging it – it is wider than the wall and therefore protrudes out. Some consider that this runs between the walls across the doorway, others that it runs around the top of the porch walls. Sitting on the frieze is a colonnade of short pillars supporting beams (blue, even though they are called golden beams, *gser gdung*) running parallel to the lower frieze. Hanging in front of the short pillars are garlands and drops (*dra ba dra phyed*, *hārārdhahāra*), depicted clearly in the 2D maṇḍala. These originally took the form of garlands of flowers hung from wooden pegs stuck into holes in stupas and similar structures. See the Sanchi image below.

Laid across the beam supported by the little pillars (and beams at the same height inside the palace) are joists (*lcam*, *phyam*) supporting a flat roofing, consisting of slats of wood and jewelled clay. There are often two or



Carving at Sanchi, clearly showing at the top, joists, flat roofing and battlemented parapet. Click image for higher resolution.

three layers of these, each protruding beyond the one below to form a cornice (*phyi tshé*). From the underside of the flat roofing hangs a fascia (*shar bu*, *bakuli*) which can take several forms. The most common (but not the only form, as is sometimes suggested) is of white water pipes. These are in the shape of upturned little bottles which would channel water from the flat roof that would collect when rain falls. Finally, on top of the flat roof is the parapet (*mda' yab*, *kramaśīrṣa*), which can also take more than one form. The most traditional is of battlement-style merlons (*btag so/stag so* ?), either flush with the edge of the flat roof or set back a little. In the Gelug tradition, the merlons are commonly in the form of half-lotus petals, flush with the edge of the roof. This form is certainly the later architectural development and can be seen on buildings such as the Red Fort in Delhi. In much older carvings, such as those at Sanchi (see again the image above right), parapets are mainly depicted in battlement-style, usually above a stepped cornice as described above. These features are represented in the third image. For clarity, the toran that should be in the background on the right has been removed.

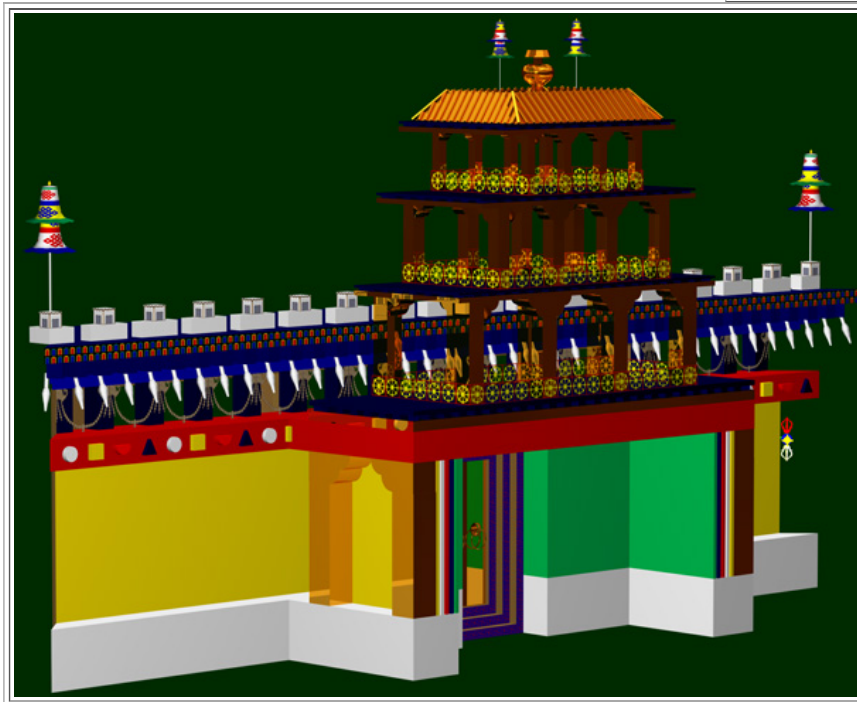
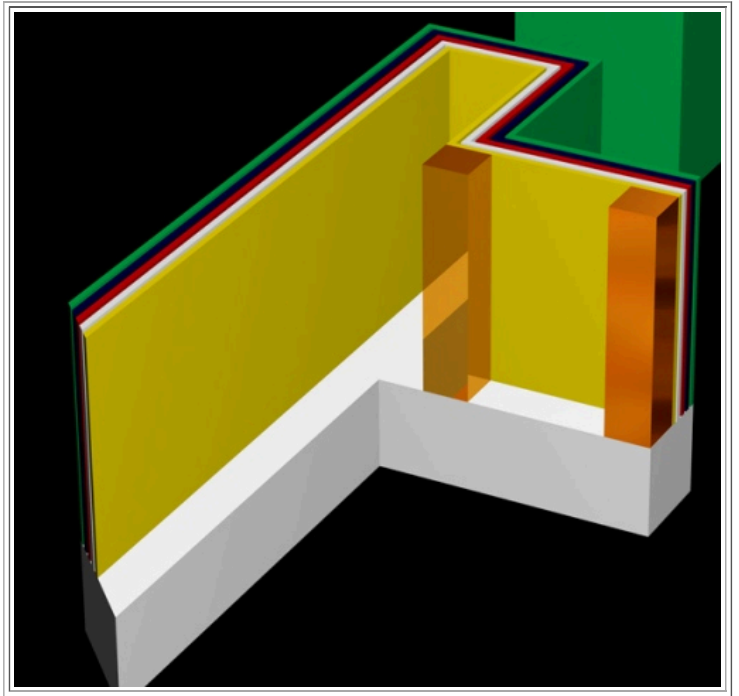
In the image at the top of the page, the structure above the porch, consisting of three stages (*rtseg ma*, *pura*) each with four pillars, is a toran (*rta babs*, *torāṇa*). This is in fact a vertical structure, rising above the porch, that is here (in the 2D maṇḍala) shown laid out horizontally. We shall return to this.

Chokle Namgyal's design

The first points made by Chokle Namgyal that differ from the usual interpretation of the 3D Kālacakra maṇḍala concern the plinth. There are three points of difference: 1) he has the plinth as a base to the walls, which is not at all unreasonable and may very well be the original intention; 2) he has the plinth running round the base of the porch, which it would logically have to do if it really were the supporting base of the walls; and, 3) his plinth is rather stout, being $\frac{1}{2}$ DU, both in width and height. This means that it protrudes beyond the bottom of the walls by the same amount as normal, but is much thicker (in height).

We then come to one of the two points that gives this page its title. Chokle Namgyal states that the height of the walls needs to be determined from the length of the pillar supporting the toran in the 2D maṇḍala. This is quite illogical, as the length of that pillar in the 2D drawing is limited by the length of the external wall of the porch-sides ($1\frac{1}{4}$ DU). To take the height of these pillars literally from the 2D drawing is no more sensible than determining the height of the parapet in the same way. That pillar is a vertical structure that cannot possibly be represented literally in a 2D

plan drawing. In the usual form of the 3D palace, the height of the walls is $2\frac{3}{4}$ DU. Given that his walls are on top of the plinth, the total height of his walls becomes $1\frac{1}{4} + \frac{1}{2} = 1\frac{3}{4}$ DU. This is quite a change, and reduces the overall height of the palace structure significantly.



He then goes on to describe the toran. This rests on the jewelled-frieze that is on top of the walls, and he points out how this fact cannot be represented in the drawn 2D maṇḍala, and that in such a maṇḍala the toran appears to sit directly on top of the walls. He makes no changes to the structure of the toran, and his design is identical to that in the normal 3D maṇḍala. We can skip a few details here, but point out that he notes that the height of the top surface of the frieze is 2 DU, the very top of the parapet is 3 DU, and that the height of the top of the 3-DU toran is therefore 5 DU. The top of the toran is taken to be the top of the flask on the top, not the parasols either side (for this reason the parasols are left out of some of the images here).

Of course, these changes to the sizing of the walls also apply to the speech and mind palaces. In addition, as a consequence of the lower structure of the body palace, the foundation (*gar bu*) inside it that supports the speech palace has to be lower, as do the foundations that support the mind palace and the Circle of Great Bliss. It is in that structure (he calls it the Circle of Awareness, *ye shes kyi 'khor lo*) that we next find Chokle Namgyal taking the size of a vertical structure in the 2D maṇḍala literally and applying it to the 3D palace.

The two images on the right show the Circle of Great Bliss in both the 2D maṇḍala and also a partly cut-away 3D palace (clicking on that image will bring up a higher resolution version). In the 2D drawing, the central feature is the main lotus on which stands Kālacakra and his consort surrounded by eight goddesses. In the corners of the square in which this lotus sits would normally be the four emblems: a tree, conch, jewel and semantron. These are not shown here. Moving outwards, next is a set of inner beams forming a square; these would normally be blue in colour, usually with a vajra design in the 2D drawing. At the outside is the set of outer beams, normally green, also with vajra designs. Between these sets of beams are pillars, 16 in total, forming a total of 16 cells (*re'u mig*) in which are lotuses.

The eight large lotuses are for four buddhas and their consorts, and the slightly smaller lotuses are for flasks.

In the 3D structure the two sets of beams, inner and outer, are supported by their own sets of pillars. (I have seen computer generated images with just one set of pillars and beams, but this is quite wrong.) The inner pillars are set the same distance apart as the outer pillars, and there is therefore a set of 12 inner (blue) pillars and a set of 16 outer (black) pillars. The latter are marked with designs of swords, jewels, wheels and lotuses. In the image shown here, the two outer pillars nearest to the camera (one sword and one lotus) have been removed. The deities are represented by their seed-characters. The whole sits on the foundation for the Circle of Great Bliss which sits in the middle of the mind palace.

The buddhas and flasks sit on the tathāgata-dais (*de gshegs 'phar ma*, tathāgatapuṭa). This term really refers to the whole platform, excluding the

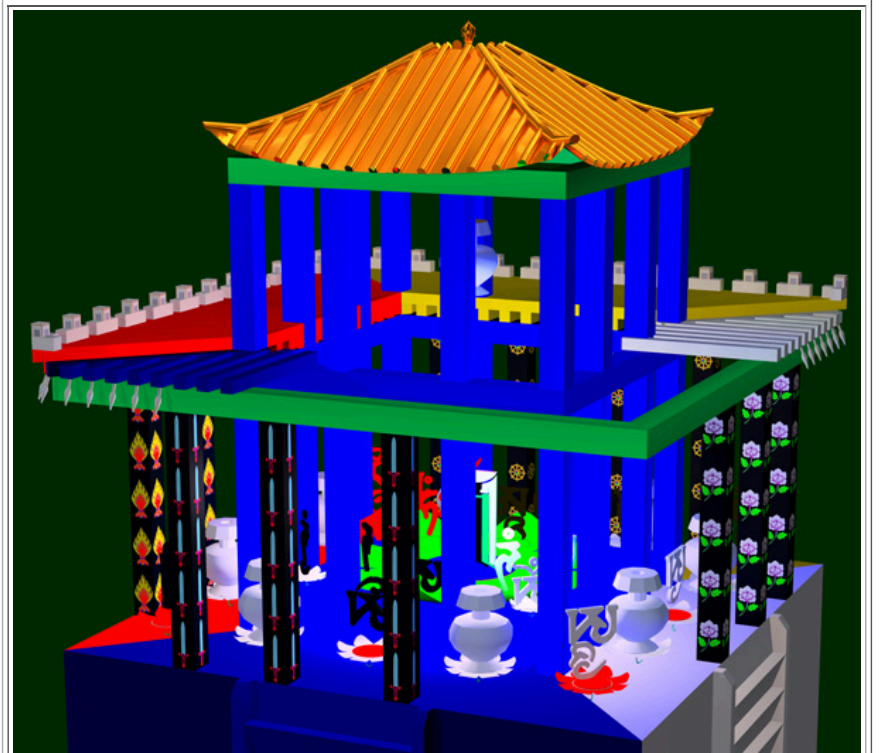
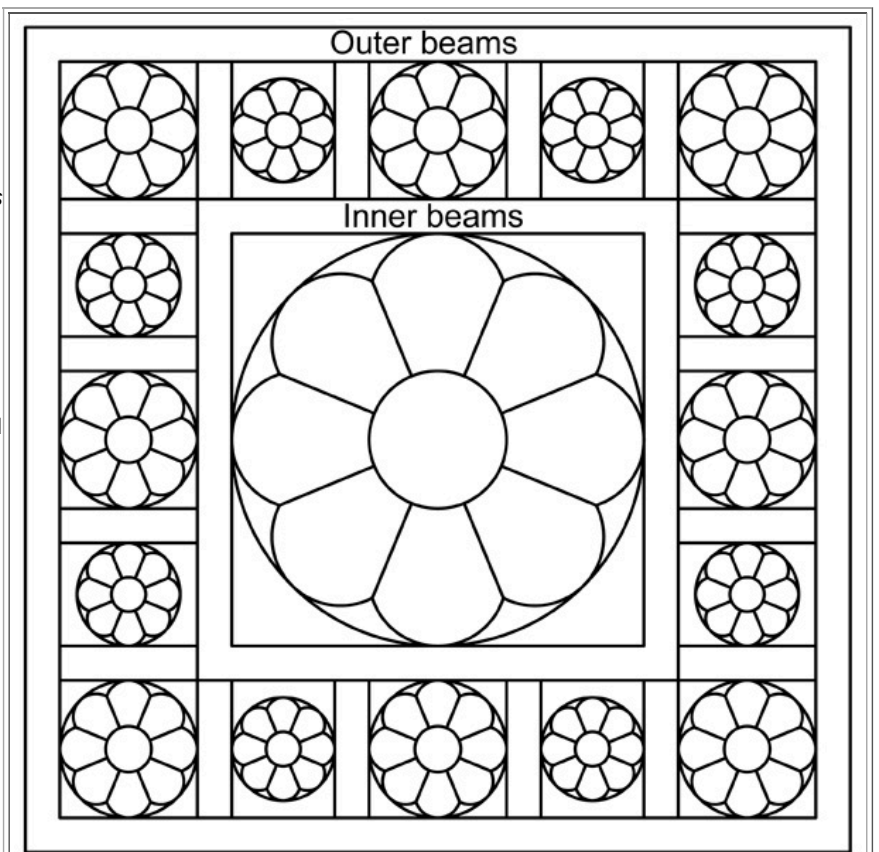
raised section for the chief deity, but usually the intention is the space between the two sets of pillars, where the tathāgatas, their consorts and the flasks have their seats. The two sets of beams on top of those pillars support joists and flat roofing to cover the tathāgata-dais; much of this nearer to the camera has been removed. (Tathāgatapuṭa is the term used in the Vimalaprabhā; in later Tibetan writings it is usually referred to as *bde gshegs 'phar ma* (Sugata-dais). As in an English language context the term tathāgata is better known than sugata (tathagata appears in the OED), it seems best to give here the Sanskrit as Tathāgatapuṭa. The word 'phar ma, puṭa, is another term for the more common deity-podium – *lha'i snam bu*, devatāpaṭṭika, such as the deity-podia in the mind, speech and body palaces.)

The chief deity site – the central section for Kālacakra and his immediate retinue – also needs protection from the elements, and so immediately above each of the 12 inner pillars, set into the flat roofing, are 12 upper pillars, also blue, that support green beams and the final top, Chinese-style, roof (*rgya phibs/phubs*). These pillars give the appearance of being extensions of the inner pillars below. The colours given here are the usual ones – there are alternatives possible. The four of these upper pillars nearest to the camera have been removed.

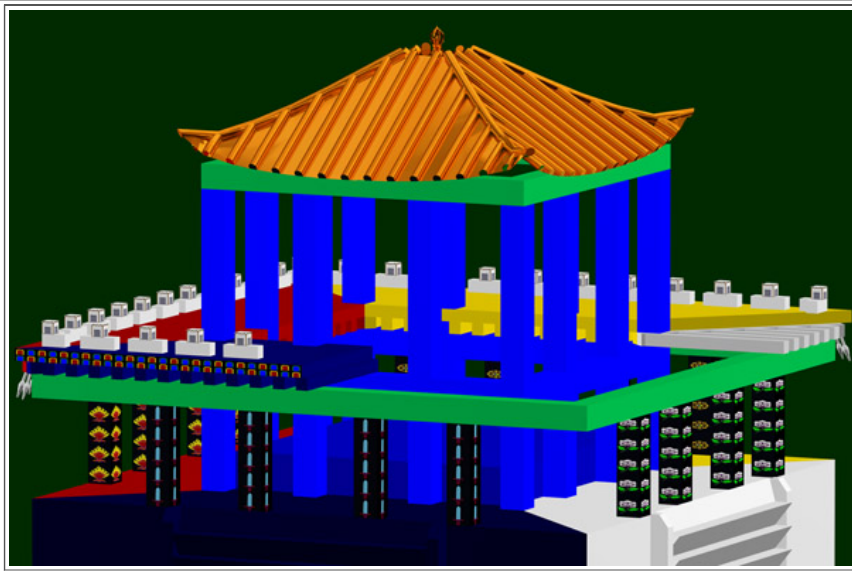
Another unit of measurement is needed here, the minor unit (mu, *cha chung*). This is in fact the unit most commonly used in describing the palace and maṇḍala, and is equal to $\frac{1}{6}$ of a door unit of the mind palace. As the inner palaces have each half the dimensions of the outer ones, if 6 mu is equal to one mind-DU, then 12 mu is one speech-DU and 24 mu is one body-DU – the latter is the unit of measurement described above.

Some care is needed here. Some writers refer to minor units that are $\frac{1}{6}$ the door units of each palace. So, one can talk of a body-mu, speech-mu as well as mind-mu. The convention suggested by Banda Gelek is that such units that are $\frac{1}{6}$ of respective door units should be called fractional units (*cha phran*). (The translation as fractional unit is suggested by the fact that Tibetan children learning about fractions in school will be discussing *cha phran*.) So, you could have a body-fu, for example. In another text, Banda Gelek states that "to take one sixth of a speech- or body-DU as a minor unit is very mistaken". At least he is consistent in that, but unfortunately, not even he always follows this convention regarding fractional units, and in at least one Kālacakra text he refers to the fractional unit as being $\frac{1}{4}$ of the respective door unit. Also, in maṇḍalas other than Kālacakra the minor unit and the fractional unit are both considered to be $\frac{1}{4}$ of a door unit. Usually. One simply needs to be very careful, and in this writer's experience it is best to convert everything to mind-mus and stick to that.

In the normal description of the palace, the two sets of pillars are 10 mu in height, and the thickness of the roofing together with the supporting beams is 2 mu. The upper pillars are given by Banda Gelek as "about 8 mu in height". The uncertainty allows for the pillars to be shorter and the Chinese-style roof taller, or vice versa, as one wishes. The important thing is that the distance from the top of the flat surface of the roof over the tathāgata-dais to the half-vajra finial (*rdo rje'i tog*) on top should be 12 mu, making the overall height of the structure 24 mu – the same as its width.



And Chokle Namgyal? See the image to the left, in which the deities are not represented. He maintains that the height of both the inner and outer sets of pillars (they have to be equal) are taken from the drawn maṇḍala. In the 2D drawing, the pillars are represented lying across the tathāgata-dais, which is only 4 mu in width. This makes the roof over the tathāgata-dais very low. He then goes on to state that Kālacakra is three times the (body) height of the buddhas and their consorts, and consequently needs three times the head-room. He therefore suggests that the upper pillars are double the height of the lower ones, at 8 mu. His logic is rather simplistic here. For a start, the buddhas are seated and Kālacakra standing, and also, floating above Kālacakra's head, below the top roof, should be one of the ten flasks (that has been removed from



the upper image to make the architecture more visible). Logically, this should make the upper pillars even taller, and even more disproportionate.

He also ignores mention of both the platform on which Kālacakra's lotus sits and the thickness of the roofing over the tathāgata-dais. It appears that he would do without the platform and the roof would be relatively thin. He is not concerned that the overall height of the structure should equal its width.

One interesting point that he makes is that the structure would look better if the outer pillars included pillars at the corners, making a total of 20 instead of 16. It is certainly an unusual concept to have such a structure lacking corner pillars, and Chokle Namgyal points out that such corner pillars would support the beams at the points where they join. Structurally, this makes a great deal of sense, and if the original intention was to have such corner pillars, they could not be represented in the drawn 2D maṇḍala. He makes a good case here, from the point of view of both style and logic. Tibetan Kālacakra experts would no doubt argue that the symbolism of the palace

requires 16 pillars and no more, but that symbolism seems to be written more from the point of view of the drawn maṇḍala (in verses quoted in the fourth chapter of the Vimalaprabhā) in which such corner pillars could not possibly be represented.

On maṇḍala architecture terminology: compared to other articles on this web site, more effort has been made here to list the Tibetan and Sanskrit equivalents of the English terms used. This is because current dictionaries are not as helpful on this subject as one would like. One problem with maṇḍala architecture is that English architectural terminology has evolved from the Greek, and often has very specific meanings, perhaps not appropriate in an Indian/Tibetan context. One therefore needs to be careful in choosing the most suitable terms. A couple of examples follow. Reading texts on western architecture suggests two possible translations for the ratnapaṭṭika (*rin chen snam bu*) that runs along the top of the walls: architrave and frieze. The latter has been chosen here because the ratnapaṭṭika is ornamented with jewels, and in both 2D and 3D maṇḍalas this ornamentation can be quite lavish. In the normal meanings of these English words, both the architrave and the frieze are part of the entablature, the frieze being the ornamented band above the architrave (which is also usually plain). Also, in the Dictionary of Architecture and Construction, Cyril Harris gives one definition of frieze as "A decorative band at or near the top of an interior wall below the cornice". This is a reasonable match for the ratnapaṭṭika, but it would be a step too far to talk in terms of an entablature in maṇḍala architecture.

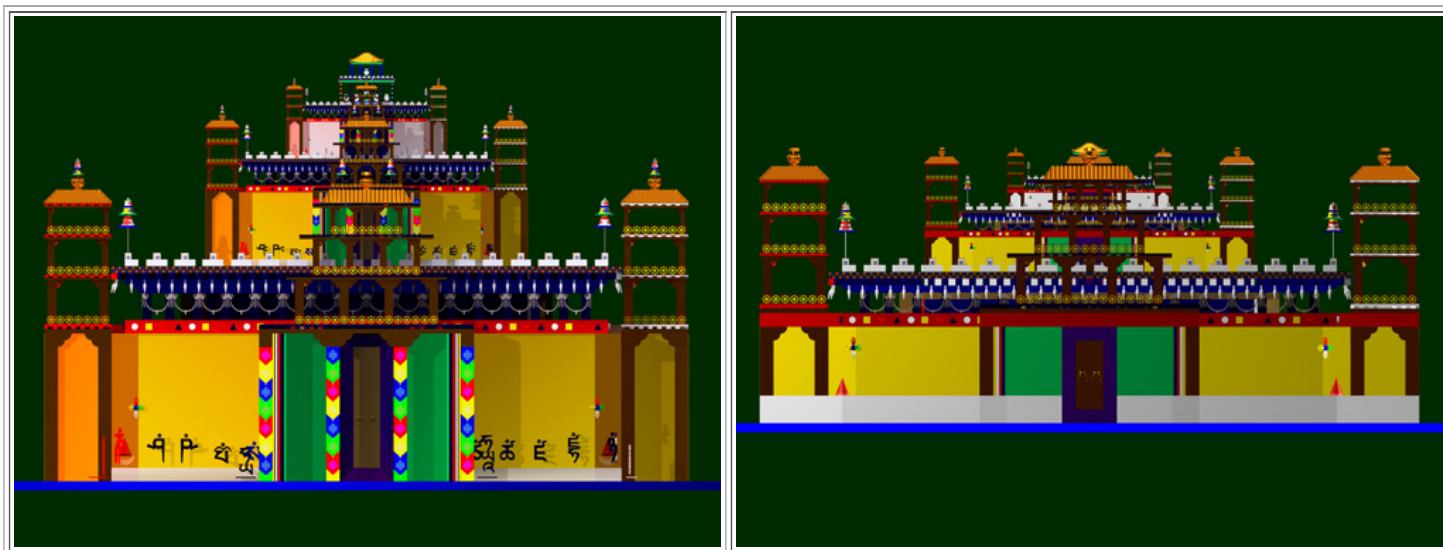
Mistakes are easily made, and in her beautiful and commendable book, "The Torana in Indian and Southeast Asian Architecture", Parul Pandya Dhar calls the horizontal elements of a toran, architraves, which seems quite inappropriate (mind you, she is not alone in this). The term used in this article, stage, avoids any unfortunate Greek connotation and is supported by the OED definition: "Each of the portions into which the height of a structure is divided; a horizontal partition." Also, unlike architrave, the term stage is compatible with the idea that people can walk or stand on these sections of a toran, as is clear in the torans at Sanchi and elsewhere, and is implied directly by the term used for the guard-rail on each stage in the Kālacakra toran. This consists of a balustrade that runs around the outside of the sets of upper pillars of the toran and completely surrounds them so as to protect anyone intoxicated from alcohol falling over the edge. For this reason they are called (literally) "drunk-protectors" (*myos pa srung ba*, *mattavāraṇa*). (Incidentally, this terminology does not apply to torans in most other non-Kālacakra maṇḍalas.) On the word torana, as Dhar correctly points out, there is no reasonable English equivalent, and as the Anglicized toran is now to be found in dictionaries such as the OED, it seems best to use that. Translations such as gateway or archway rather miss the point and are somewhat misleading.

The overall impact of the differences in the design of the maṇḍala palace as described by Chokle Namgyal can be seen in the orthogonal views below. The one on the left is the more normal structure, following the descriptions by Tāranātha and Banda Gelek. That on the right is the design by Chokle Namgyal.

One point that is often made regarding the maṇḍala palace is that its height and width should be the same. The width that is taken is the distance between the internal surfaces of opposite walls of the body palace, and in the normal design, this is precisely equal to the height of the tip of the vajra-finial on the top of the roof (196 mind-mu). Chokle Namgyal describes his structure in terms of cubits (*khru*), where 1 body-DU equals 50 cubits. The internal width of the body palace is 400 cubits (8 DU, as is normal), but he gives the overall height of his structure as 268 cubits. He states that this does not quite meet the normal criterion that the overall height and width should be the same. Only a little different then!

He points out the two main strengths of his design as follows: the tips of all the torans are at exactly the same height, and there is no doubt that this helps make his palace very visually attractive; also, he writes that going from one level to another requires much less climbing of stairs than normal and his design avoids the problem of having too many stairs (the tops of the stairs are visible on the foundation for the Circle of Great Bliss in a couple of the images above). He also adds that in his design, the top of the flat surface of a roof of an outer palace is flush with the top of the plinth of the inner one, and this makes the overall structure more attractive. Reading his text suggests that aesthetics were a major driver of his ideas. He is clearly convinced that the maṇḍala palace of Kālacakra must be a very beautiful structure.

Chokle Namgyal is not alone in interpreting vertical sizes literally from the 2D maṇḍala. Others, including one of his more famous students, also consider that the height of the toran pillars should be taken from the 2D drawing, and instead of lowering the walls, places the toran in front of the porch, almost blocking it. This will be illustrated on this web site at a later date.



(The present article should be considered a work-in-progress, as the text and images may well be updated in the future. In particular the overall height of the 3D maṇḍala palace model shown here is a little less than that given by Chokle Namgyal, and the reason for this is not yet clear. Some other details could also do with improvement or clarification.)



E. Henning.
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